



PH 608: Computer assisted quantum mechanics

Course Code	PH 608
Course Name	Computer assisted quantum mechanics
L-T-P-C	2-0-3-3
Prerequisites	PH301 / PH513 (Quantum Mechanics), PH524 / EP403 (Physics of Atoms and Molecules), PH613 (Computational methods for physicists)/EP302 (Computational methods for engineering)
Intended for	UG/PG/I-PhD/PhD elective
Mutual Exclusion	NA
Approval	50th BoA (Jul 2023)

Course Contents

- **The single-particle problem:** Time independent Schrodinger equation and its solution with Numerov's method, Bound state solutions for one-dimensional (1D) case such as Harmonic oscillator, Schrodinger equation for central potentials, solutions of hydrogen atom, scattering from different type of central potentials, Response of atoms to external fields (20 hours)
- **Variational method:** The variational principle, Numerical solutions to quantum mechanical problems using variational methods, Plane-wave basis set and Non-orthonormal basis set (15 hours)
- **Multi-electron systems:** Basics of Hartree-Fock (HF) methods and its numerical implementation to a few selected problems, going beyond HF methods, density functional theory and its implementation with some specific examples for simple atomic systems. (15 hours)
- **Time propagation:** Spectral methods, direct numerical integration, split operator and Crank-Nicolson methods. Implementation of these methods to a few quantum mechanical systems (20 hours)

Text Books

1. J. M. Thijssen, **Computational Physics**, Cambridge University Press, 2007.
2. J. Izaac and J. Wang, **Computational Quantum Mechanics**, Springer, 2018.

References

1. Paolo Giannozzi, **Numerical methods in quantum mechanics**. (Online lecture notes, <http://www.fisica.uniud.it/~giannozz/Corsi/MQ/LectureNotes/mq.pdf>)
2. R. H. Landau, M. J. Páez and C. C. Bordeianu, **Computational physics**, WILEY-VCH Verlag, 2015.